

Lecture Slide – 6
Determinants and
Cramer's Rule

Determinants

- If a matrix is square (that is, if it has the same number of rows as columns), then we can assign to it a number called its determinant.
 - *Determinants can be used to solve systems of linear equations-- as we will see later in the section.*
 - *They are also useful in determining whether a matrix has an inverse.*

Determinant of 1×1 matrix

- We denote the determinant of a square matrix A by the symbol $\det(A)$ or $|A|$.
- We first define $\det(A)$ for the simplest cases.
 - If $A = [a]$ is a 1×1 matrix, then $\det(A) = a$.

Determinant of a 2×2 Matrix

The determinant of the 2×2 matrix is:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of an $n \times n$ Matrix

- To define the concept of determinant for an arbitrary $n \times n$ matrix, we need the following terminology.

Let A be an $n \times n$ matrix:

- The minor M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i_{th} row and j_{th} column of A .
- The cofactor A_{ij} of the element a_{ij} is:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Determinant of an $n \times n$ Matrix

- For example, A is the matrix
$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{bmatrix}$$
- The minor M_{12} is the determinant of the matrix obtained by deleting the first row and second column from A .

$$\begin{aligned} M_{12} &= \begin{vmatrix} \cancel{2} & \cancel{3} & \cancel{1} \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} & M_{33} &= \begin{vmatrix} 2 & 3 & \cancel{-1} \\ 0 & 2 & 4 \\ \cancel{-2} & \cancel{5} & \cancel{6} \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = 2 \cdot 2 - 3 \cdot 0 = 4 \\ &= 0(6) - 4(-2) = 8 \end{aligned}$$

- So, the **cofactor** $A_{12} = (-1)^{1+2} M_{12} = -8$
- So, $A_{33} = (-1)^{3+3} M_{33} = 4$

Determinant of an $n \times n$ Matrix

- Note that the cofactor of a_{ij} is simply the minor of a_{ij} multiplied by either 1 or -1 , depending on whether $i + j$ is even or odd.
 - Thus, in a 3×3 matrix, we obtain the cofactor of any element by prefixing its minor with the sign obtained from the following checkerboard pattern.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Determinant of a Square Matrix

- If A is an $n \times n$ matrix, the determinant of A is obtained by multiplying each element of the first row by its cofactor, and then adding the results.

$$\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$
$$= a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n}$$

Determinant of a 3×3 Matrix

Example-1: Evaluate the determinant of the matrix.

$$\det(A)$$

$$= \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 2 & 4 \\ 5 & 6 \end{vmatrix} - 3 \begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} + (-1) \begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix}$$

$$= 2(2 \cdot 6 - 4 \cdot 5) - 3[0 \cdot 6 - 4(-2)] - [0 \cdot 5 - 2(-2)]$$

$$= -16 - 24 - 4$$

$$= -44$$

Expanding the Determinant

- In our definition of the determinant, we used the cofactors of elements **in the first row only**.
 - This is called expanding the determinant by the first row.
 - In fact, we can expand the determinant by any row or column in the same way, and obtain the same result in each case.
 - We won't prove this, though.

Expanding the Determinant

Example 1. (a)

- Expanding by the second row, we get:

$$\begin{aligned}\det(A) &= \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & -4 \\ -2 & 5 & 6 \end{vmatrix} \\ &= -0 \begin{vmatrix} 3 & -1 \\ 5 & 6 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ -2 & 6 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} \\ &= 0 + 2[2 \cdot 6 - (-1)(-2)] - 4[2 \cdot 5 - 3(-2)] \\ &= 0 + 20 - 64 = -44\end{aligned}$$

Expanding the Determinant

Example 1.(b)

- Expanding by the **third column**, we get:

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= -1 \begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} + 6 \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix}$$

$$= -[0 \cdot 5 - 2(-2)] - 4[2 \cdot 5 - 3(-2)] + 6[2 \cdot 2 - 3 \cdot 0]$$

$$= -4 - 64 + 24 = -44$$

In both cases, we obtain the same value for the determinant as when we expanded by the first row.

Invertibility Criterion

- If A is a square matrix, then A has an inverse if and only if $\det(A) \neq 0$.
 - We will not prove this fact.
 - However, from the formula for the inverse of a 2×2 matrix, you can see why it is true in the 2×2 case.

Invertibility Criterion

- Show that the matrix A is not invertible.
 - We begin by calculating the determinant of A .
 - Since all but one of the elements of the second row is zero, we expand the determinant by the second row.

$$A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{bmatrix}$$

Determinant to Show Matrix Is Not Invertible

If we do so, we see from this equation that only the cofactor A_{24} needs to be calculated.

$$\begin{aligned}\det(A) &= \begin{vmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{vmatrix} \\ &= -0 \cdot A_{21} + 0 \cdot A_{22} - 0 \cdot A_{23} + 3 \cdot A_{24} \\ &= 3A_{24}\end{aligned}$$
$$\begin{aligned}&= 3 \begin{vmatrix} 1 & 2 & 0 \\ 5 & 6 & 2 \\ 2 & 4 & 0 \end{vmatrix} \\ &= 3(-2) \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \\ &= 3(-2)(1 \cdot 4 - 2 \cdot 2) \\ &= 0\end{aligned}$$

Since the determinant of A is zero, A cannot have an inverse—by the Invertibility Criterion.

Row and Column Transformations

- The preceding example shows that, if we expand a determinant about a row or column **that contains many zeros**, our work is reduced considerably.
- We don't have to evaluate the cofactors of the elements that are zero.
- This principle often simplifies the process of finding a determinant by introducing zeros into it without changing its value.
- If A is a square matrix, and if the matrix B is obtained from A by adding a multiple of one row to another, or a multiple of one column to another, then

$$\det(A) = \det(B)$$

Row and Column Transformations

- Find the determinant of the matrix A .

$$A = \begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 24 & 6 & 1 & -12 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

- Does it have an inverse?

Row and Column Transformations

- If we add -3 times row 1 to row 3, we change all but one element of row 3 to zeros:

$$\begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 0 & 0 & 4 & 0 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

- This new matrix has the same determinant as A .

Row and Column Transformations

- If we expand its determinant by the third row, we get:

$$\det(A) = 4 \begin{vmatrix} 8 & 2 & -4 \\ 3 & 5 & 11 \\ 2 & 2 & -1 \end{vmatrix}$$

- Now, adding 2 times column 3 to column 1 in this determinant gives us the following result.

Row and Column Transformations

$$\begin{aligned}\det(A) &= 4 \begin{vmatrix} 0 & 2 & -4 \\ 25 & 5 & 11 \\ 0 & 2 & -1 \end{vmatrix} = 4(-25) \begin{vmatrix} 2 & -4 \\ 2 & -1 \end{vmatrix} \\ &= 4(-25)[2(-1) - (-4)2] \\ &= -600\end{aligned}$$

- Since the determinant of A is not zero, A does have an inverse.

Systems of Linear Equations

A system of linear equations can be presented in different forms

$$\left. \begin{array}{rcl} 2x_1 + 4x_2 - 3x_3 & = & 3 \\ 2.5x_1 - x_2 + 3x_3 & = & 5 \\ x_1 & & -6x_3 = 7 \end{array} \right\} \Leftrightarrow \begin{bmatrix} 2 & 4 & -3 \\ 2.5 & -1 & 3 \\ 1 & 0 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$$

Standard form

Matrix form

$$|\mathbf{A}| \cdot |\mathbf{x}| = |\mathbf{b}|$$

Cramer's Rule

Linear Equations and Determinants

- The solutions of linear equations can sometimes be expressed using determinants.
- To illustrate, let's solve the following pair of linear equations for the variable x .

$$\begin{cases} ax + by = r \\ cx + dy = s \end{cases}$$

Linear Equations and Determinants

- To eliminate the variable y , we multiply the first equation by d and the second by b , and subtract.

$$adx + bdy = rd$$

$$bcx + bdy = bs$$

$$adx - bcx = rd - bs$$

Linear Equations and Determinants

- Factoring the left-hand side, we get:

$$(ad - bc)x = rd - bs$$

- Assuming that $ad - bc \neq 0$, we can now solve this equation for x :

$$x = \frac{rd - bs}{ad - bc}$$

- Similarly, we find:

$$y = \frac{as - cr}{ad - bc}$$

Linear Equations and Determinants

- The numerator and denominator of the fractions for x and y are determinants of 2×2 matrices.
- So, we can express the solution of the system using determinants as follows.

Cramer's Rule for Systems in Two Variables

- The linear system
$$\begin{cases} ax + by = r \\ cx + dy = s \end{cases}$$

- has the solution

$$x = \frac{\begin{vmatrix} r & b \\ s & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} \qquad y = \frac{\begin{vmatrix} a & r \\ c & s \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

- provided

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$$

Cramer's Rule for Systems in Two Variables

- Using the notation for determinants

$$D = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad D_x = \begin{bmatrix} r & b \\ s & d \end{bmatrix} \quad D_y = \begin{bmatrix} a & r \\ c & s \end{bmatrix}$$

- the solution of the system can be written as:

$$x = \frac{|D_x|}{|D|} \quad \text{and} \quad y = \frac{|D_y|}{|D|}$$

Cramer's Rule for Systems in Two Variables

- Use Cramer's Rule to solve the system.

$$\begin{cases} 2x + 6y = -1 \\ x + 8y = 2 \end{cases}$$

Cramer's Rule for Systems in Two Variables

- For this system, we have:

$$|D| = \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = 2 \cdot 8 - 6 \cdot 1 = 10$$

$$|D_x| = \begin{vmatrix} -1 & 6 \\ 2 & 8 \end{vmatrix} = (-1)8 - 6 \cdot 2 = -20$$

$$|D_y| = \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} = 2 \cdot 2 - (-1)1 = 5$$

Cramer's Rule for Systems in Two Variables

- The solution is:

$$x = \frac{|D_x|}{|D|} = \frac{-20}{10} = -2$$

$$y = \frac{|D_y|}{|D|} = \frac{5}{10} = \frac{1}{2}$$

Cramer's Rule

- Cramer's Rule can be extended to apply to any system of n linear equations in n variables in which the determinant of the coefficient matrix is not zero.
- As we saw in the preceding section, any such system can be written in matrix form as:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Cramer's Rule

- By analogy with our derivation of Cramer's Rule in the case of two equations in two unknowns, we let:
 - D be the coefficient matrix in this system.
 - D_{x_i} be the matrix obtained by replacing the i_{th} column of D by the numbers $b_1, b_2, \dots, b_n$ that appear to the right of the equal sign.
- The solution of the system is then given by the following rule.

Cramer's Rule

- Suppose a system of n linear equations in the n variables $x_1, x_2, \dots, x_n$ is equivalent to the matrix equation $DX = B$, and $|D| \neq 0$.
 - Then, its solutions are:

$$x_1 = \frac{|D_{x_1}|}{|D|}, x_2 = \frac{|D_{x_2}|}{|D|}, \dots, x_n = \frac{|D_{x_n}|}{|D|}$$

where D_{x_i} is the matrix obtained by replacing the i_{th} column of D by the $n \times 1$ matrix B .

Cramer's Rule for a System of Three Variables

- Use Cramer's Rule to solve the system.

$$\begin{cases} 2x - 3y + 4z = 1 \\ x \quad \quad + 6z = 0 \\ 3x - 2y \quad \quad = 5 \end{cases}$$

- First, we evaluate the determinants that appear in Cramer's Rule.

Cramer's Rule for a System of Three Variables

$ D = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ 3 & -2 & 0 \end{vmatrix} = -38$	$ D_x = \begin{vmatrix} 1 & -3 & 4 \\ 0 & 0 & 6 \\ 5 & -2 & 0 \end{vmatrix} = -78$
$ D_y = \begin{vmatrix} 2 & 1 & 4 \\ 1 & 0 & 6 \\ 3 & 5 & 0 \end{vmatrix} = -22$	$ D_z = \begin{vmatrix} 2 & -3 & 1 \\ 1 & 0 & 0 \\ 3 & -2 & 5 \end{vmatrix} = 13$

- Note that D is the coefficient matrix and that D_x , D_y , and D_z are obtained by replacing the first, second, and third columns of D by the constant terms.

Cramer's Rule for a System of Three Variables

- Now, we use Cramer's Rule to get the solution:

$$x = \frac{|D_x|}{|D|} = \frac{-78}{-38} = \frac{39}{19}$$

$$y = \frac{|D_y|}{|D|} = \frac{-22}{-38} = \frac{11}{19}$$

$$z = \frac{|D_z|}{|D|} = \frac{13}{-38} = -\frac{13}{38}$$

Limitations of Cramer's Rule

- In systems with more than three equations, evaluating the various determinants involved is usually a long and tedious task.
- Moreover, the rule doesn't apply if $|D| = 0$ or if D is not a square matrix.